CT DOSE AI · ADVANCED TRACK

Where deep learning earns its place

Three extensions beyond the physics model: a pre-scan predictor (built), virtual dose reduction (planned), and a scout-image network (research). With the data each one needs.

> **The premise.** The dose law is physics, so no network will beat it — that question is settled, and the report shows the evidence. The open problems are the ones physics cannot answer: *what will the scanner choose for this patient*, *how low can the dose go before the image stops being diagnostic*, and *what does this particular body do to the beam*. All three are learnable, and the third one needs images.

Track	Question it answers	Data needed	Status
1 Pre-scan	What dose will *this patient* receive, before the scan?	the 50 exams (done)	**built**
2 Virtual dose reduction	How low can we go, without irradiating anyone to find out?	raw/DICOM images, 30–50 exams	planned
3 Scout network	What does this body do to the beam?	localizer images, 300+ exams	research

1 · Pre-scan dose prediction — built

1.1 The gap it closes

The dose model needs the mAs. But the tube-current modulation chooses the mAs only when the patient is already on the table, so before the scan the most useful question — *what dose will this patient receive?* — could not be answered at all. A second, purely statistical stage now answers it:

Stage 1 (learned): BMI → mAs the ATCM will choose, BMI → scan length
Stage 2 (physics): mAs, kV, length → $\mathrm{CTDI_{vol}}$, DLP

1.2 Here the flexible models win

Same competition, same repeated grouped cross-validation as the dose models — but this time there is no formula to fall back on, and the ranking inverts:

Model (BMI as the only input)	mAs error	Scan-length error
Median, ignoring BMI (baseline)	27.9%	9.6%
Log-linear in BMI	23.4%	5.6%
Log-quadratic in BMI	21.4%	**5.4% (selected)**
Gaussian process	21.2%	5.3%
k-nearest neighbours	21.4%	5.9%
Random forest	**19.5% (selected)**	5.4%
Gradient boosting	20.6%	5.7%
Neural network	22.8%	6.1%

The random forest beats the linear model by 4 percentage points and the no-BMI baseline by 8. That is a real, measured gain from a flexible model — the honest counterpart to the dose result, where the flexible models lost.

1.3 Honest uncertainty

Chaining two models means chaining two errors. The app convolves the out-of-fold residual distributions of both stages (they are independent) rather than adding variances under a normality assumption:

Example patient (120 kV)	mAs	$\mathrm{CTDI_{vol}}$ (mGy)	DLP (mGy·cm)
BMI 22	466 (309–741)	28.2 (18.7–42.1)	650 (420–1027)
BMI 27	658 (437–1048)	39.8 (26.5–59.5)	919 (593–1452)
BMI 35	1040 (690–1655)	62.9 (41.8–93.9)	1452 (937–2294)

Roughly $\pm$35% instead of the $\pm$7% available once the mAs is known. Wide — but that width *is* the information: it says how unpredictable the modulation is, and it is exactly the quantity a department needs when consenting a patient or forecasting a year's collective dose.

> **Where it is in the app.** A new tab, **Pre-scan (BMI only)**: expected mAs, $\mathrm{CTDI_{vol}}$ and DLP with intervals, the two learned curves against the historical exams, and the model competition in an expander. Four new automated tests cover it, including one that requires the pre-scan interval to be at least twice the width of the fixed-protocol interval — so the extra uncertainty can never be quietly lost.

2 · Virtual dose reduction — the highest-value next step

2.1 The idea

The clinical bandit learns from radiologist scores, one patient at a time. That is safe but slow: 300 patients to learn what 300 labels can teach. **Noise insertion** breaks that limit. Starting from an already-acquired full-dose scan, one can synthesise the image that *would have been obtained* at 70%, 50% or 30% of the dose, by adding correlated noise in the projection (or image) domain according to the scanner's noise model.

> Nobody is irradiated. One existing exam yields four or five labelled examples at different dose levels, and the radiologist scores images of the *same patient* — so the comparison is paired, which removes most of the between-patient variance and multiplies statistical power.

2.2 Method

1. **Noise model.** Quantum noise in the projection domain is Poisson in the detected photon count N. To simulate dose factor $f < 1$: $\tilde{N} = \mathrm{Binomial}(N, f)$, or in log domain add Gaussian noise of variance $\sigma^2 = \frac{1-f}{f} \cdot \frac{1}{N}$, then add the electronic-noise term that dominates at low N and is what makes very low kV fail. Validate the simulator against two or three real low-dose phantom scans before trusting it on patients.
2. **Where.** Projection-domain insertion (needs raw/CT-projection data from the vendor) is the gold standard; image-domain insertion with a noise-power-spectrum-matched texture is the practical fallback when raw data is not exportable and is well described in the literature.
3. **Blinded reading.** Each patient's images at 100/70/50/30% dose are anonymised, randomly ordered and scored 1–4 by two readers. Inter-reader agreement (κ) is reported.
4. **Fit.** The same Bayesian logistic acceptability model already in the app, now with hundreds of paired labels; the paired structure is handled with a per-patient random effect.
5. **Result.** The dose level at which $P(\text{acceptable}) = 0.95$, per BMI band — measured, on your own images, with your own readers, before changing a single patient protocol.

2.3 What to ask for

2.4 Effort and output

Roughly 3–6 weeks of work: simulator (1–2 weeks), validation on phantom (a few days), reading sessions (scheduled), analysis (1 week). It produces a defensible, site-specific low-dose protocol — and it is a publishable study in its own right.

Item	Detail	Who provides
Images	30–50 consecutive CCTA studies, DICOM, anonymised	radiology / PACS
Raw data (ideal)	vendor projection data for the same studies, if exportable	vendor / physicist
Dose reports	the RDSR of each study (gives true scan length too)	PACS
Reading time	2 readers $\times$ $\sim$4 hours for 150–200 blinded image sets	radiologists
Approval	ethics / clinical-audit sign-off for retrospective image reuse	department

3 · Scout-image network — the research track

3.1 Why an image beats BMI

BMI is a scalar summary of a 3-D body. Two patients with BMI 30 can differ by a factor of two in chest attenuation. The scanner already acquires the information that matters — the **scout** / **topogram** / **localizer**, a low-dose projection radiograph used for planning — and it is exactly the input the modulation algorithm itself uses. A convolutional network on the scout can therefore predict what BMI cannot:

Output	Why it matters	Label source
Water-equivalent diameter	enables SSDE — patient dose, not phantom dose	computed from the axial images (AAPM 220)
mAs the ATCM will select	replaces the 19.5% BMI model; expect single digits	the exam itself
Expected image noise	the missing half of the optimisation	measured in a fixed ROI
Organ / effective dose	risk, not console metrics	Monte-Carlo simulation as the teacher

3.2 Architecture and training plan

- **Input:** the AP and lateral scout images, resized to 512×256, plus the patient's age, sex and BMI as tabular side-inputs.
- **Backbone:** a small pretrained CNN (ResNet-18 or EfficientNet-B0, ImageNet weights) — pretraining matters enormously at these sample sizes — with the tabular inputs concatenated before the head.
- **Heads:** multi-task regression (water-equivalent diameter, mAs, noise), which regularises far better than three separate models.
- **Physics-informed loss:** predict $\log$ mAs and add a consistency term so that the implied $\mathrm{CTDI}_{\mathrm{vol}}$ obeys the fitted scanner law — the network then cannot contradict the physics it sits on top of. This is the point at which the word "physics-informed neural network" becomes technically correct for this project.
- **Validation:** patient-level splits, never image-level; the same repeated cross-validation discipline; the BMI model as the baseline to beat.
- **Data:** 300 exams to show a signal, 1000+ for a solid result. Augmentation (translation, mild scaling, intensity jitter) and heavy weight decay throughout.

3.3 Deep surrogate for Monte-Carlo dose (stretch goal)

A full Monte-Carlo organ-dose simulation takes minutes to hours per patient. A network trained on a few thousand simulated cases reproduces it in milliseconds, which turns organ dose from a research exercise into something the app can show while the patient is still in the room. This is a genuine use of deep learning as a *function approximator for an expensive simulator* — the setting where it is unambiguously the right tool.

> **Sequencing matters.** Track 3 needs images, an ethics pathway and several hundred exams. Track 2 needs images from a few dozen exams and produces a clinical result on its own. Track 1 is already running on the data you have. Do them in that order, and each one funds the next with evidence.

4 · The single data request

Everything above reduces to one email to the department. In priority order:

1. **RDSR (radiation dose structured reports)** for the existing 50 exams and, going forward, automatically. Gives $\mathrm{CTDI}_{\mathrm{vol}}$, DLP, kV, mAs *and the true scan length* per series — it settles the "SCAN SIZE CM" question and

removes all manual typing.

2. **The protocol fields:** gating mode (prospective/retrospective), collimation, pitch, rotation time, reconstruction kernel and iterative strength, heart rate.
3. **A diagnostic-quality score (1–4)** recorded with the report, even informally. This is the label that makes every learning loop work.
4. **Anonymised DICOM images** for 30–50 exams, including the scout, for tracks 2 and 3.
5. **More exams:** 150–300 per exam type is the point where the models stop being limited by sample size.

> **What each request buys.** RDSR $\rightarrow$ DLP error from 8% towards 1–2%. Protocol fields $\rightarrow$ removes the last of the $\mathrm{CTDI_{vol}}$ error and lets the tool reason about gating. Quality scores $\rightarrow$ the clinical learning loop starts working. Images $\rightarrow$ virtual dose reduction and the scout network. More exams $\rightarrow$ the model competition can finally be won by something flexible.